For systems in which the memory kernel decays fairly rapidly, it is not unreasonable to consider its approximation by a linear combination of exponential functions, a so-called *Prony series*:

$$\gamma(t) = \sum_{k=1}^{k_{\max}} \frac{c_k}{\tau_k} \exp\left[-\frac{t}{\tau_k}\right], \quad t \geq 0. \tag{8.60}$$

The properties of the system in the limit $\tau_k \to 0$ are studied in [293].

The use of this representation for the memory kernel allows us to replace the history-dependent equations (8.57)–(8.58) by a Markovian system in a larger dimensional space, by introducing k additional degrees of freedom per physical degree of freedom. This method obviously increases storage requirements, and, in large systems, it may be significant. However, the evolution of these additional variables is elementary and the overall computational cost low.

8.7.1 Using an Extended Variable Representation

We introduce a set of extended variables whose evolution is described by standard Brownian motions using Ornstein-Uhlenbeck (OU) processes. For this treatment, let the particle positions and momenta be enumerated as (q_i, p_i), $i = 1, \ldots, \mathrm{N_c}$ and define the extended variable, $\eta_{i,k}(t)$, associated with the kth term in the Prony series in interaction with the ith physical variable by

$$\eta_{i,k}(t) = -\int_0^t \frac{c_k}{\tau_k} \exp\left[-\frac{(t-s)}{\tau_k}\right] p_i(s)\mathrm{d}s. \tag{8.61}$$

Equations (8.57)–(8.58) can then be rewritten in the form

$$\mathrm{d}q_i = m_i^{-1} p_i \mathrm{d}t \tag{8.62}$$

$$\mathrm{d}p_i = F_i \mathrm{d}t + \sum_{k=1}^{k_{\max}} \eta_{i,k}\mathrm{d}t + \mathrm{R}_i \mathrm{d}t. \tag{8.63}$$

We note that $\eta_{i,k}$ is the solution of an ODE of the form

$$\mathrm{d}\eta_{i,k} = -\frac{1}{\tau_k}\eta_{i,k}(t)\mathrm{d}t - \frac{c_k}{\tau_k} p_i \mathrm{d}t. \tag{8.64}$$

We have thus resolved the convolution integral.